

Professional & Engaging Speaker Notes for Project Video Walkthrough

Opening (0:00 – 0:45)

[Camera on, warm but confident tone]

"Hi everyone. I'm [Your Name], and this is my project for the IDEAS Summer Internship Program. Today, we're going to explore two very different worlds using data. First, we'll travel back in time to 1880 and track global temperature changes all the way to 2016. Then, we'll switch gears completely and teach a computer to recognize handwritten digits – without ever giving it the answers. Sound interesting? Let's jump into the notebook."

Part 1: Global Temperature Time-Series Analysis – A 136-Year Climate Journey

Cell 1 – Loading the Data (0:45 – 1:30)

[Start screen share, point to code]

"We start by importing pandas and NumPy – the bread and butter of data analysis in Python. Then we load our temperature dataset. Let's run this."

Run cell, show output.

"Here's what we're working with: two temperature sources – GCAG from NOAA and GISTEMP from NASA. A date column, currently just text. And the 'Mean' column – that's the temperature anomaly, meaning how much warmer or cooler that month was compared to a long-term baseline. Notice the dates are backwards – the newest is on top. We'll fix that right now."

Cell 2 – Converting and Sorting Time (1:30 – 2:15)

"We convert the Date column into a real datetime object. Then we sort from oldest to newest. Watch this transformation."

Run cell, show output.

"Boom. Now we're in 1880. The earliest records. Every trend we discover from now on rests on this solid chronological foundation. That's the secret to good time-series analysis."

Cell 3 – Data Inspection (2:15 – 2:45)

"Before we get carried away, let's check the data's health. Shape? 3,288 rows. And missing values? Zero. Perfect. No need to guess or fill anything."

Cell 4 – Extracting Year and Month (2:45 – 3:20)

"We want to see patterns by year and by month. So we pull the year and month numbers out of the date column. Simple, but incredibly useful."

Run cell, show output.

"Now every temperature reading comes with its own year and month label. Ready for grouping."

Cell 5 – 12-Month Moving Average – Seeing Through the Noise (3:20 – 4:30)

"Raw monthly temperatures are noisy – seasons, random weather events. To see the real long-term trend, we calculate a 12-month moving average. It's like applying a smoothing filter. I do this separately for each source using a rolling window of 12 months."

Run cell, scroll to tail.

"Look at the last five rows of 2016. The moving average for GCAG in October – 0.9819. For GISTEMP in November – over 1.017. These numbers are higher than the raw monthly anomalies. Why? Because the moving average carries the warmth of the previous months forward. The heat isn't just spiking – it's accumulating."

Cell 6 – Focusing on the Last 20 Years (4:30 – 5:15)

"We don't need all 136 years for every question. Let's zoom in on the most dramatic period – the last 20 years, from 1997 to 2016. I create a new DataFrame called df_last20."

Run cell, show shape.

"480 rows. That's 20 years × 12 months × 2 sources. A perfect little window into recent climate change."

Cell 7 – Which Source Is Warmer on Average? (5:15 – 6:00)

"Let's ask a simple but powerful question: Over the entire record, which source shows a higher average anomaly – GCAG or GISTEMP?"

Run cell.

"GCAG says 0.0488 degrees. GISTEMP says 0.0244 degrees. Both positive – meaning the planet is indeed warmer than the baseline. The small difference comes from how each organization handles data gaps. But the direction is clear and consistent."

Cell 8 – Year by Year – The Staircase of Warming (6:00 – 7:00)

"Now for the real evidence. I group by year and compute the average anomaly for each year. Let me show you the last five."

Run cell.

"2012: 0.6295. 2013: 0.6618. 2014: 0.7421. 2015: 0.8824. 2016: 0.9644. That's not random. That's a staircase. And the steps get steeper each year. This is the global warming signal – persistent, accelerating, and unmistakable."

Cell 9 – The Hottest Month? Prepare to Be Surprised (7:00 – 8:00)

"Here's a fun twist. If I asked you which month is warmest on average, you'd probably say July or August. Let's check the data."

Run cell.

"October. 0.0526 degrees. September is second. July and August are lower. Why? Because oceans absorb heat slowly and release it with a delay, pushing the peak warmth into autumn. This is a great example of how data can challenge our everyday assumptions."

Cell 10 – The Pivot Table – A Heatmap of Two Decades (8:00 – 9:30)

"Finally, we build a pivot table. Years go down the rows, months across the columns. Each cell holds the average temperature anomaly for that specific year and month. This is the crown jewel of our analysis."

Run cell, scroll slowly.

"Look at the late 1990s – values mostly in the 0.3 to 0.6 range. Now scroll down to 2015 and 2016. February 2016 – 1.271 degrees. October 2015 – 1.025. November – 1.002. December – 1.115. The entire matrix lights up. This isn't just summer heat. Every month, every season has shifted upward. That is the signature of a changing climate."

Part 2: Unsupervised Learning – Can a Machine Learn Without a Teacher?

Cell 11 – Loading MNIST Digits (9:30 – 10:15)

"Now we completely change direction. We move from climate science to machine learning. I load the digits dataset from scikit-learn – a smaller version of the famous MNIST handwritten digits. It contains 1,797 images of digits 0 through 9. Each image is 8×8 pixels, flattened into a list of 64 numbers. I split into features X and true labels y. The true labels are only for evaluation – the machine will not see them during clustering."

Run cell, show output.

"1,797 samples, 64 features each. Ready for unsupervised learning."

Cell 12 – K-Means Clustering (10:15 – 11:15)

"I import KMeans and set the number of clusters to 10 – because we know there are ten digits. I use a fixed random state so the results are reproducible. Then I fit the model and predict cluster labels in one step."

Run cell.

"We get an array of cluster labels, one per image. But here's the catch: K-Means assigns arbitrary numbers. Cluster 0 might actually be the digit 7. So we need to map them."

Cell 13 – Evaluating with F1 Score (11:15 – 12:30)

"I wrote a function that, for each cluster, finds the most frequent true digit inside it. Then it maps all cluster labels accordingly. After mapping, we compute the macro-averaged F1 score – a balanced measure of accuracy across all digits."

Run cell.

"The result: 0.8628 – about 86.3%. For an unsupervised algorithm that never saw a single label, that's excellent. It means K-Means successfully groups visually similar digits together. The mistakes happen where digits look alike – like 1 and 7, or 3 and 8. But overall, the machine learned the structure of handwritten digits all by itself."

Closing (12:30 – 13:15)

[Stop screen share, look at camera]

"So there you have it. Two datasets, two completely different problems, one notebook. The temperature analysis showed a clear, accelerating warming trend – with 2016 as the hottest year and autumn months surprisingly the warmest. The clustering experiment showed that even without labels, a simple algorithm can discover meaningful patterns in handwritten digits. Both stories come from the same tool – data analysis with Python. All code is on my GitHub, link in the report. Thank you for watching – I hope you enjoyed the journey as much as I did."